

Lab Instructions: Multivariate Linear

March, 12th, 2026

Objective

The goal of this lab is to build a multivariate linear regression model to analyze and predict the progress of diabetes using the diabetes dataset. The dataset consists of 10 features and one target variable, which measures the diabetes progression.

Steps

1. Import Libraries and Dataset

In this step, you will import the necessary libraries and load the diabetes dataset.

- Import the required libraries such as NumPy, pandas, Matplotlib, Seaborn, and scikit-learn.
- Load the diabetes dataset using the `load_diabetes()` function from scikit-learn.
- Convert the dataset into a pandas DataFrame and check the first few rows of the dataset.

Expected Outcome: Display the first five rows of the dataset.

Grading Point 1: Show the first five rows of the diabetes dataset.

2. Visualize and Display Data Distributions

Use Seaborn's `pairplot` to visualize the distribution and relationships between features in the dataset.

- Use the `pairplot()` function from Seaborn to create a pairplot of all features.

- Observe the relationships between different features visually.

Expected Outcome: Display a pairplot showing the relationships between features.

Grading Point 2: Visualize the pairwise relationships among features using a pairplot.

3. Explore Relationships between Features and Target Variable

For each feature, create a scatter plot to show the relationship between the feature and the target variable (diabetes progression).

- For each feature in the dataset, create a scatter plot to visualize its relationship with the target variable.
- Label the axes appropriately and provide meaningful titles.

Expected Outcome: Display scatter plots showing the relationship between each feature and the target variable.

Grading Point 3: Show scatter plots for each feature against the target variable.

4. Feature Selection and Data Splitting

Split the dataset into training and testing sets.

- Split the data into training and testing sets using an 80-20 or 70-30 ratio.
- Print the shape of the training and testing sets.

Expected Outcome: Display the shapes of the training and testing sets.

Grading Point 4: Show the shapes of the training and testing sets after splitting the data.

5. Build and Train the Multivariate Linear Regression Model

In this step, you will create and train the linear regression model using the training data.

- Build a linear regression model using scikit-learn's `LinearRegression()`.
- Train the model on the training dataset.

- Output the model's intercept and coefficients.

Expected Outcome: Display the intercept and coefficients of the model.

Grading Point 5: Show the model's intercept and coefficients.

6. Prediction and Model Evaluation

Use the trained model to predict the target variable for the test set and evaluate the model's performance.

- Make predictions using the test set.
- Calculate the Mean Squared Error (MSE) and the R^2 score for the model.

Expected Outcome: Display the Mean Squared Error (MSE) and the R^2 score.

Grading Point 6: Show the evaluation results: MSE and R^2 score.

7. Residuals Analysis

To assess the model's fit, analyze the residuals.

- Calculate the residuals (difference between the actual and predicted values).
- Plot the residuals using a histogram with a kernel density estimate (KDE).

Expected Outcome: Display the residuals distribution.

Grading Point 7: Show the residuals distribution with a histogram and KDE plot.

8. Visualize the Regression Line

For one of the features, plot the regression line against the test set to visualize the fit.

- Choose one feature (e.g., **age**) and plot both the actual and predicted target values.
- Overlay the regression line on the scatter plot.

Expected Outcome: Show the regression line for a single feature.

Grading Point 8: Visualize a feature with the regression line.

9. Feature Importance Analysis

Examine the importance of each feature in the model.

- Display the regression coefficients for each feature in a sorted manner.

Expected Outcome: Show the importance of each feature based on its regression coefficient.

Grading Point 9: Show the features sorted by their coefficients.

10. Cross-Validation

To assess the model's generalization ability, perform cross-validation.

- Use 5-fold cross-validation to evaluate the model's performance across different folds.
- Report the cross-validation scores and the mean score.

Expected Outcome: Display the cross-validation results.

Grading Point 10: Show the results of cross-validation and the mean score.